∞-brush: Controllable Large Image Synthesis with Diffusion Models in Infinite Dimensions

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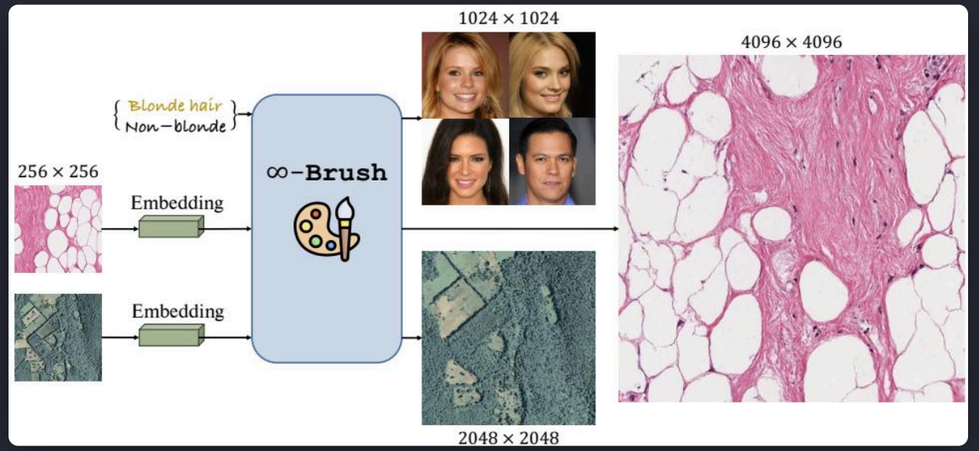
Date: August 2025

Problem Statement

- Current diffusion models limited to fixed training resolutions
- Cannot exceed 1024×1024 without fidelity loss (e.g., SDXL)
- Patch-based methods fail to capture long-range spatial relationships
- Heavy reliance on local information during generation
- No existing model enables controllable generation at 4096×4096
- Computational demands scale quadratically with resolution

Our Solution

- Introduce ∞-Brush: First conditional diffusion model in function space
- Generates images at arbitrary resolutions up to 4096×4096
- Key innovation: Cross-attention neural operator
- Enables conditioning in infinite-dimensional space
- Training efficiency: Only 0.4% pixel subsets per image
- $65,536$ pixels instead of 16 million for 4096×4096

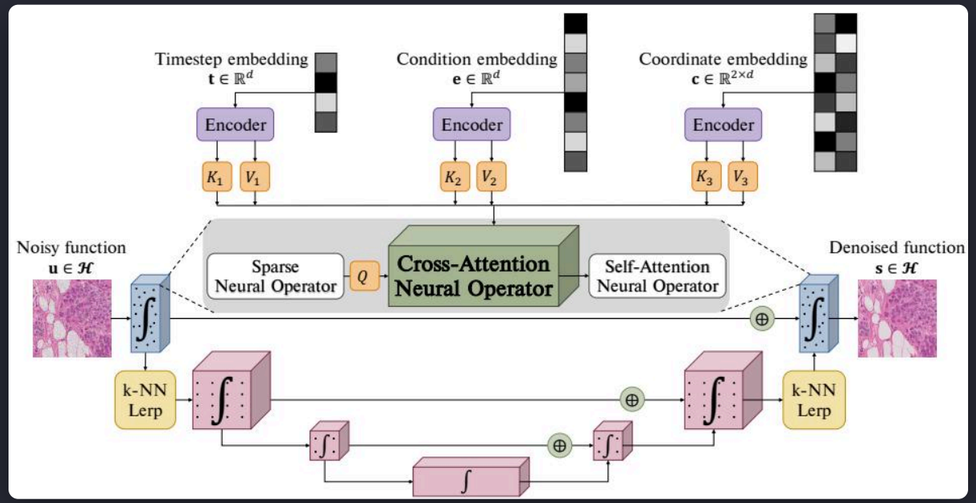


Related Work Limitations

- **Finite-dimensional models** (SDXL, Matryoshka)
- Fixed resolution constraint: Cannot exceed training resolution
- Quadratic computational cost with resolution
- **Patch-based methods** (MultiDiffusion, Graikos et al.)
- Captures local details but loses global structure
- No conditioning for controllable generation
- **Infinite-dimensional models** (∞-Diff)
- Enables arbitrary resolution but no conditional control

Core Innovation

- **Cross-attention neural operator**
- Replaces vanilla attention in function space
- Complexity: $O((N + \Sigma N_l)d^2)$ vs. $O(N^2 d)$
- **Conditional diffusion process**
- Forward: $Q(u_t|u_0) = N(u_t; \sqrt{\bar{\alpha}_t}Au_0, (1-\bar{\alpha}_t)ACA^T)$
- **Function-space intuition**: an image is a function over continuous coordinates, not a fixed-size grid — the model learns from sparse samples yet generates at any resolution
- **vs. finite-dim & patch-based**: grid models are capped at training resolution and patches condition only locally; conditioning the whole function enables controllable arbitrary-resolution synthesis
- **Training efficiency**
- 0.4% pixel subsets per image ($65,536$ vs. 16M pixels)

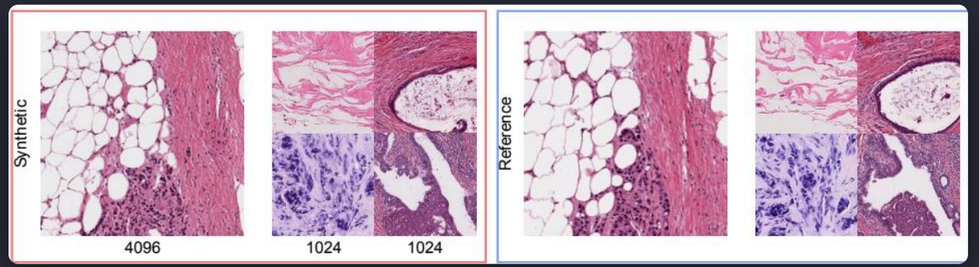


Experimental Setup

- **Datasets & conditioning type**
- CelebA-HQ (1024×1024): 30k faces, attribute conditioning (blonde vs. non-blonde hair)
- TCGA-BRCA (4096×4096): 57k histopathology patches, Quilt self-supervised embedding conditioning
- NAIP satellite (1024×1024): 35k images, DINO self-supervised embedding conditioning
- ∞-Diff baseline is unconditional; ∞-Brush adds conditional control
- **Training vs. evaluation resolutions**
- Training: $65,536$ pixels ($\sim 0.4\%$) sampled per image, independent of target resolution
- Evaluation: full-resolution CLIP FID (global) + 256×256 crop FID (local detail)
- **Baselines**
- SDXL (finite-dim), Graikos et al. (patch-based), ∞-Diff (unconditional)

Results: Global Structure

- **Global structure (CLIP FID)**
- ∞-Brush clearly outperforms patch-based Graikos et al. at 4096×4096 (TCGA-BRCA)
- ∞-Brush leads SDXL and Graikos et al. at 1024×1024 (NAIP, BRCA 5×)
- SDXL cannot be trained at 4096×4096, so no comparison at that scale
- **Local detail (Crop FID)**
- Modest trade-off: fine detail slightly behind patch-based methods
- **Qualitative**
- Generated 4096×4096 images preserve the global structure of reference images
- Takeaway: superior global structure at resolutions no baseline can reach



Results: Efficiency & Trade-offs

- **Efficiency**
- Constant training cost across resolutions; SDXL infeasible at 4096×4096
- **Global vs. local trade-offs**
- vs. SDXL: better global structure at 1024×1024, modest trade-off in fine detail
- vs. patch-based: better global coherence at both resolutions, slightly worse local detail
- vs. ∞-Diff: adds conditional control with better global structure
- **Ablation**
- Removing the cross-attention operator substantially harms both global and local metrics
- Takeaway: consistent scalability across resolutions

Key Contributions

- First conditional diffusion model in function space
- Enables controllable generation at arbitrary resolutions
- Cross-attention neural operator for infinite dimensions
- Linear complexity vs. quadratic for vanilla attention
- Training on `0.4%` pixel subsets
- Achieves `4096 × 4096` synthesis with 10x lower memory

Limitations & Future Work

- **Limitations**
- Fine details slightly worse than patch-based methods
- Smaller parameter count (450M vs. 860M for [10])
- **Future work**
- Design local neural operators for fine details
- Transfer knowledge from finite-dimensional models
- Scale to higher resolutions (8192×8192)

Conclusion

- ∞-Brush overcomes resolution limits of existing models
- First to enable controllable 4096×4096 synthesis
- Key innovation: Cross-attention neural operator
- Maintains global structure while preserving details
- Computational efficiency:
- 12 hr/epoch vs. 140 hr for SDXL at 1024×1024
- Constant training cost across resolutions